

JOURNAL OF CHEMICAL INFORMATION AND MODELING

Submission Cover Letter and Author Checklist

The *Journal of Chemical Information and Modeling* now requires authors to complete and upload this form in lieu of the Cover Letter in the submission process. Note: An editorial was published on guidelines for publishing papers in *JCIM: J. Chem. Inf. Model.* 2015, 55, 719. [DOI: 10.1021/acs.jcim.5b00180](https://doi.org/10.1021/acs.jcim.5b00180). Submissions should follow these guidelines.

General Manuscript Information

Title: To probe the binding of CX-5461, an anti-cancer DNA G-quadruplex stabilizer, to human telomeric, cKIT-1, and c-Myc G-quadruplexes and a DNA duplex using molecular dynamics binding simulations

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Reason for submission to the *Journal of Chemical Information and Modeling* (explain importance to field and major findings): Stabilizing DNA G-quadruplexes (G4s) of cancerous cells has become a novel therapeutic technique that has been shown to inhibit cancerous cell growth and replication leading to the cancerous cells death. A promising new anti-cancer medication, CX-5461 is in Phase I/II clinical trials for advanced hematologic malignancies and cancers with BRCA1/2 deficiencies. Experimental evidence identified CX-5461's specific roles at G4s as the ability to bind to and stabilize G4 structures in human cancer cell lines, increase the number of G4 sites, as well as selectivity bind to G4 structures on the human genome. These properties are extremely advantageous for cancer therapeutics however the detailed binding mode and mechanism for which CX-5461 interacts with the G4 structures remains elusive. Our analyses sought to characterize the binding of CX-5461 to the human telomeric, c-KIT1, c-Myc G4s as well as a DNA duplex to provide insight into the binding process and help to explain the experimentally reported order of binding induced stability.

Additional information (e.g., if the paper was considered by another journal explain what happened and provide previous critique if available) or special instructions (e.g., publish back-to-back with companion paper, reviewer exclusions):

Data Requirements

All Supporting Information files containing information on molecular structure and associated property data must be in a machine readable format for import into typical modeling packages.

Brief Checklist (see the editorial for more details [Editorial: Method and Data Sharing and Reproducibility of Scientific Results](#))

Does your manuscript contain the “**Data and Software Availability**” section?

Yes – make sure it clearly discusses how all data and software used can be obtained.

Make sure the **Data and Software Availability** section describes how all data can be obtained and discusses why some data needs to be proprietary. If all data is proprietary the manuscript will be sent back without review.

Make sure the **Data and Software Availability** section describes how all software can be obtained. If commercial software, please describe how the software can be obtained for review purposes in the cover letter. Make sure the manuscript describes the theory at a level that allows a person skilled in the art could implement the method. If the software is unavailable the manuscript will be returned without review.

No - the manuscript will be sent back without review.

Does your manuscript report new biological activity of compounds?

- ☐ **Yes** – Please provide sufficient evidence to support your report:
1. For the most interesting compounds, (e.g., most drug like, etc.) dose-response curves or other evidence should be reported.
 2. Please provide evidence to support your report that these compounds are compelling. This could include counter screens or other kinds of tests to show that they are well behaved.
 3. Provide chemical structures of all compounds whose activity you are reporting (as SMILES in the SI).
- ☐ **Not applicable**

Have you done the necessary experimental validation work to ensure your compound(s) is (are) not a bad actor(s) (e.g., PAINS, aggregator, etc.)? Participating ACS journals have published an editorial on this topic ([DOI: 10.1021/acs.icim.7b00105](https://doi.org/10.1021/acs.icim.7b00105)) that should be carefully considered and followed.

- ☐ **Yes** – Please explain what you have done to validate your compounds.
- ☐ **No** – please note your manuscript will be evaluated editorially before being sent out for review and if the conclusion is reached that this issue hasn't been addressed satisfactorily the manuscript will be returned to you.
- ☐ **Not applicable**

Author Checklist

Prior to submission, please confirm by marking all relevant boxes.

Preparation

- ☐ Table of Contents graphic is included as the last page of the manuscript or uploaded as a separate file designated "Graphic for manuscript." Label as "Table of Contents graphic."
- ☐ Figures, charts, tables, schemes, and equations should be embedded in the text. You may supply separate graphics at revision. You may provide references in any style, but they must be complete including titles.

Submission

- ☐ The "previous submissions" custom question must be accurately answered.
- ☐ All authors are included in Step 2 in the order they appear on the manuscript and contain at least as much information as appears on the manuscript.
- ☐ Simplified Molecular Input Line Entry System ([SMILES](#)) are provided for compounds presented in the manuscript. Upload the .csv file in Supporting Information and clearly labeled as "SMILES". For reference, please review *J Med Chem*'s guidance on [Molecular Formula Strings](#).

Prior Publication

The *Journal of Chemical Information and Modeling* authors are allowed to deposit an initial draft of their manuscript in a preprint service such as ChemRxiv, arXiv or bioRxiv. Please note any use of a preprint server, and as appropriate, state how the manuscript has been adjusted/updated between deposition and submission. All other prior/redundant publication is forbidden. Upon publication in the *Journal of Chemical Information and Modeling*, authors are advised to add a link from the preprint to the published paper via the Digital Object Identifier (DOI).

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